



Log Analyzer

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A Python-based tool for analyzing Linux
auth.log security events

Python Fundamentals Project - XE105

Why This Tool Was Built

This tool was built to simplify the analysis of Linux authentication logs. The `auth.log` file contains important security information, but reading it manually can be difficult and time-consuming. The Log Analyzer extracts key events such as `sudo` commands, failed `sudo` attempts, `su` usage, user creation, user deletion, and password changes. By organizing this data into a clear report, the tool helps identify system activity and potential security issues more efficiently.

Main goals:

- Extract important authentication events from `auth.log`
- Highlight suspicious activity such as failed `sudo` attempts
- Present the results in a clear and readable terminal report

How the Script Works

The script receives a Linux auth.log file as input and analyzes it line by line. Each log entry is checked against predefined patterns to identify important authentication and security events.

When a relevant event is found, the script extracts key details such as the timestamp, executing user, command, target user, action, and event information. The results are then organized into clear report sections and displayed in the terminal with colored output for better readability.

Process Flow:

auth.log file



Read each log line



Detect event type



Extract important fields



Print structured report

The script does not modify the log file. It only reads and analyzes the data.

Sudo Command Detection

The first section of the report shows successful sudo command usage. For each detected sudo event, the script extracts the timestamp, executing user, and command.

This helps identify which users executed privileged commands and when those commands were performed.

== SUDO COMMANDS ==

```
1. 2026-06-07T09:00:01.100000-04:00 | User: kali | Command: /usr/bin/apt update
2. 2026-06-07T09:02:11.200000-04:00 | User: kali | Command: /usr/bin/whoami
3. 2026-06-07T09:16:28.900000-04:00 | User: root | Command: /usr/sbin/userdel testloguser
4. 2026-06-07T09:18:39.010000-04:00 | User: root | Command: /usr/sbin/useradd sudoaddeduser
5. Jun 7 09:20:50 | User: kali | Command: /usr/bin/ls /root
```

Failed Sudo Attempts

Failed sudo attempts are highlighted as security alerts.

The script marks these events with ALERT! and displays the timestamp, user, attempted command, and failure reason.

This makes suspicious authentication activity easier to detect during log review.

== FAILED SUDO ATTEMPTS ==

```
1. ALERT! 2026-06-07T09:04:22.300000-04:00 | User: kali | Command: /usr/bin/id | Reason:  
1 incorrect password attempt
```

su Usage Monitoring

The script also detects su session activity.

It extracts the timestamp, source user, source UID, target user, target UID, and session action.

This helps track when users switched to another account, especially when switching to root.

== SU USAGE ==

```
1. 2026-06-07T09:06:33.400000-04:00 | User: kali | UID: 1000 | Target: root |  
   Target UID: 0 | Action: session opened | Details: 2026-06-07T09:06:33.400000  
-04:00 kali su[21464]: pam_unix(su:session): session opened for user root(uid  
=0) by kali(uid=1000)  
2. 2026-06-07T09:08:44.500000-04:00 | User: Unknown user | UID: Not available  
   | Target: root | Target UID: Not available | Action: session closed | Detail  
s: 2026-06-07T09:08:44.500000-04:00 kali su[21464]: pam_unix(su:session): ses  
sion closed for user root
```

User Management Events

The analyzer monitors user account management activity, including newly created users and deleted users. It detects both direct user management log entries and user management commands executed through sudo.

This is useful for identifying account changes that may affect system security.

== NEW USERS ==

```
1. 2026-06-07T09:10:55.600000-04:00 | Username: testloguser | Details: 2026-06-07T09:10:55.600000-04:00 kali useradd[1204]: new user: name=testloguser, UID=1001, GID=1001, home=/home/testloguser, shell=/bin/bash
2. 2026-06-07T09:18:39.010000-04:00 | Username: sudoaddeduser | Details: 2026-06-07T09:18:39.010000-04:00 kali sudo: root : TTY=pts/1 ; PWD=/root ; USER=root ; COMMAND=/usr/sbin/useradd sudoaddeduser
```

== DELETED USERS ==

```
1. 2026-06-07T09:14:17.800000-04:00 | Username: oldloguser | Details: 2026-06-07T09:14:17.800000-04:00 kali userdel[1220]: delete user 'oldloguser'
2. 2026-06-07T09:16:28.900000-04:00 | Username: testloguser | Details: 2026-06-07T09:16:28.900000-04:00 kali sudo: root : TTY=pts/1 ; PWD=/root ; USER=root ; COMMAND=/usr/sbin/userdel testloguser
```

Password Changes and Summary

Password change events are extracted and displayed with their timestamp and affected username.

The summary section provides a quick overview of all detected events, making the final report easier to review and understand.

== PASSWORD CHANGES ==

1. 2026-06-07T09:12:06.700000-04:00 | Username: **testloguser** | Details: 2026-06-07T09:12:06.700000-04:00 kali passwd[1244]: pam_unix(passwd:chauthtok): password changed for testloguser
2. Jun 7 09:22:01 | Username: **legacyuser** | Details: Jun 7 09:22:01 kali passwd[1300]: pam_unix(passwd:chauthtok): password changed for legacyuser

== SUMMARY ==

Sudo commands: 5

Failed sudo attempts: 1

Su usage events: 2

New users: 2

Deleted users: 2

Password changes: 2

Conclusion

The Python Log Analyzer successfully converts raw Linux authentication logs into a structured and readable security report. It detects privileged command usage, failed sudo attempts, su activity, user account changes, and password changes. The tool was tested with Kali-style auth.log entries and provides a clear summary for quick security review.

Final result:

- All required authentication events were detected
- Output is readable and color-highlighted
- The log file is analyzed in read-only mode